EE3621: Digital Signal Processing

Lecture 17: Short-Time Fourier Transform (STFT) & Spectrogram
(III B.Tech EEE)

Lecture Notes (40-Minute Plan)

1 Lecture Plan & Objectives (40 Minutes Breakdown)

- **00:00 – 05:00 (5 mins): Motivation: Non-stationary signals**. Why DFT/DTFT fail for time-varying signals.
- **05:00 – 12:00 (7 mins): STFT Definition & Physical Intuition**. Math formulation and local spectrum interpretation.
- **12:00 – 17:00 (5 mins): Window Function Role**. Rectangular vs. Hanning/Hamming windows, spectral leakage.
- **17:00 – 25:00 (8 mins): Time-Frequency Resolution Trade-off**. Heisenberg uncertainty principle.
- **25:00 – 30:00 (5 mins): Spectrogram**. Definition, parameters, and speech formants example.
- **30:00 – 34:00 (4 mins): Overlap-Add (OLA) Method**. Perfect reconstruction condition and proof.
- **34:00 – 37:00 (3 mins): Advanced Views**. Wigner-Ville Distribution and DFT-bank interpretation.
- **37:00 – 40:00 (3 mins): Applications & Checkpoints**. Radar, ECG, Audio compression. 3 Checkpoint questions.

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2 Motivation: Non-stationary Signals

The Discrete-Time Fourier Transform (DTFT) and the Discrete Fourier Transform (DFT) are powerful tools. The DTFT is defined as:

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \tag{1}$$

The Stationarity Assumption: The underlying assumption of the DFT and DTFT is that the signal $x[n]$ is **stationary**. This means its statistical properties and frequency content do not change over time. When we compute the standard Fourier transform, we sum over all time. The resulting spectrum $X(e^{j\omega})$ tells us *what* frequencies exist, but it destroys any information about *when* those frequencies occurred.

Real-world Non-stationary Signals:

- **Speech:** Your vocal tract changes shape. A vowel sound has different harmonics than a consonant.
- **EEG (Brainwaves):** Brain electrical activity changes depending on the state of the subject.
- **Audio:** A piano playing a melody produces different notes at different times.

3 STFT Definition

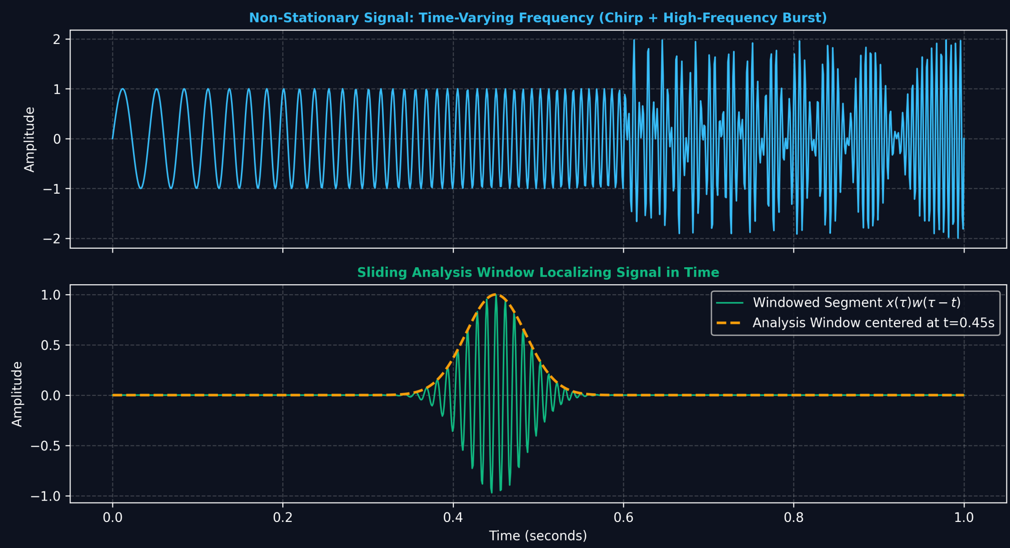

Figure 1: Short-Time Fourier Transform: Sliding Temporal Window Localizing Time-Varying Spectral Content.

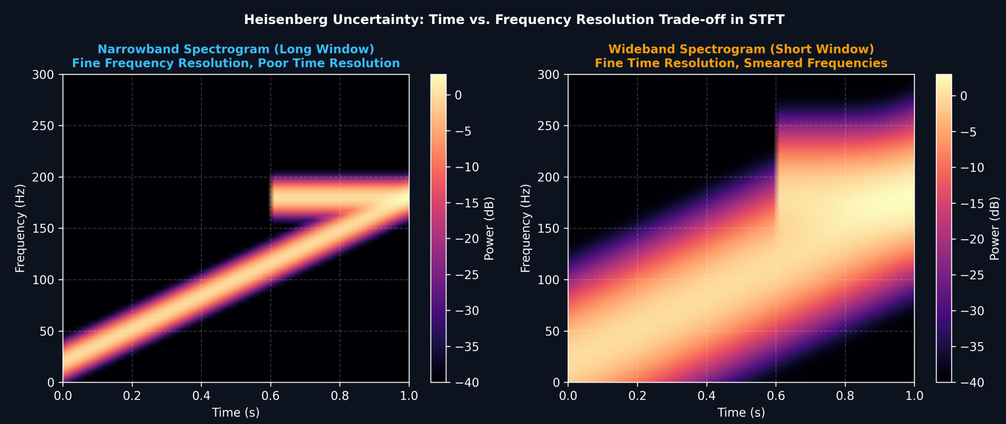

Figure 2: Spectrogram Resolution Trade-off: Narrowband (Fine Frequency) vs. Wideband (Fine Time) Representation.

To analyze a signal at a specific time n, we multiply the signal by a window function $w[m-n]$ centered at time n. This window isolates a short segment of the signal around time n. We then take the Fourier transform of this isolated segment.

The Short-Time Fourier Transform (STFT) is defined as:

$$X[n,k] = \sum_{m=-\infty}^{\infty} x[m]w[m-n]e^{-j\frac{2\pi}{N}km} \tag{2}$$

Step-by-Step Breakdown:

1. $x[m]$ is our infinite-length signal.
2. $w[m-n]$ is the **analysis window** of length M, shifted to time n.
3. $x[m]w[m-n]$ is the windowed signal.
4. The summation is the N-point DFT of the windowed signal.
5. The result $X[n,k]$ represents the **local spectrum** of the signal at time n.

4 The Role of the Window Function

The choice of the window function $w[m]$ is crucial.

The Rectangular Window:

$$w_{rect}[m] = \begin{cases} 1, & 0 \leq m \leq M-1 \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

Because the rectangular window has sharp transitions, its frequency response has a narrow mainlobe but very high sidelobes. This causes **spectral leakage** (Gibbs phenomenon), where energy from one frequency obscures nearby frequencies.

Hamming Window: To reduce spectral leakage, we taper the window:

$$w_{hamming}[m] = 0.54 - 0.46\cos\left(\frac{2\pi m}{M-1}\right) \quad \text{for } 0 \leq m \leq M-1 \tag{4}$$

Tapering reduces sidelobes (less leakage) but widens the mainlobe (decreased frequency resolution).

5 Time-Frequency Resolution Trade-off

The **Heisenberg Uncertainty Principle** states that for continuous signals:

$$\Delta t \cdot \Delta f \geq \frac{1}{4\pi} \tag{5}$$

For discrete signals, a similar bound limits the bandwidth-duration product:

- **Wide Window (Large M):** Excellent frequency resolution, poor time resolution.
- **Narrow Window (Small M):** Excellent time resolution, poor frequency resolution.

You cannot simultaneously achieve arbitrarily high resolution in both time and frequency.

6 Spectrogram

The **Spectrogram** visualizes the STFT as a 2D magnitude squared plot:

$$\text{Spectrogram}(n,k) = |X[n,k]|^2 \tag{6}$$

Practical parameters include window length (e.g., 20-30 ms for speech), hop size (overlap), and FFT size. In a speech spectrogram, vowels appear as dark bands of energy called **formants**.

7 Overlap-Add (OLA) Method & Perfect Reconstruction

To convert an STFT back to the time domain, we use the Overlap-Add method:

$$x[n] = \frac{\sum_{r=-\infty}^{\infty} y_r[n]}{\sum_{r=-\infty}^{\infty} w[n - rH]} \tag{7}$$

Theorem: Perfect reconstruction requires $\sum_r w[n - rH] = C$ (a constant).
Proof:

1. Let $y_r[n] = x[n]w[n - rH]$ be the inverse FFT of frame r.
2. Summing these overlapping frames: $\sum_r y_r[n] = \sum_r x[n]w[n - rH] = x[n] \sum_r w[n - rH]$.
3. If $\sum_r w[n - rH] = C$, then $\sum_r y_r[n] = Cx[n] \implies x[n] = \frac{1}{C} \sum_r y_r[n]$.

8 Alternative Views & Applications

Wigner-Ville Distribution: Optimal resolution but introduces artificial cross-terms for multi-component signals.

DFT-Bank Interpretation: STFT can be viewed as passing the signal through a uniform bank of modulated bandpass filters.

Applications: Speech processing (formants), MP3 audio compression (using MDCT), radar Doppler processing, and ECG analysis.

9 Key Formulas

Concept	Formula
STFT	$X[n, k] = \sum_m x[m]w[m - n]e^{-j\frac{2\pi}{N}km}$
Spectrogram	$S(n, k) = \lvert X[n, k]\rvert^2$
Uncertainty Principle	$\Delta t \cdot \Delta f \geq \frac{1}{4\pi}$
OLA Reconstruction	$x[n] = \frac{\sum_r y_r[n]}{\sum_r w[n-rH]}$

10 Checkpoint Questions

1. **Q1:** If you increase the window length M, what happens to time and frequency resolution?
   Answer: Increasing M spans a longer duration, worsening time resolution but improving frequency resolution (narrower mainlobe).
2. **Q2:** Why prefer a Hamming window over rectangular?
   Answer: Rectangular windows cause severe spectral leakage due to sharp edges. Hamming windows taper smoothly, reducing sidelobes and leakage.
3. **Q3:** What is the maximum hop size H for a rectangular window $w[n] = 1$ $(0 \leq n \leq 99)$ to satisfy COLA without division?
   Answer: $H = 100$. The frames must tile perfectly without gaps or overlapping accumulations to maintain a constant sum of 1.